

What changed between the ML4PS submission and the arXiv version

The body is the same paper. **No number in Table 1 or Figure 1 changed, the headline count is still 7 of 10, and the reference list is the same.** What the four-page limit had cut is now written out: four new appendices, four extended ones, and six additions to the body. The last pages show the body with every change marked — **red = removed**, **blue = added** — and stop where the body stops. The appendices are not marked: four of them are new in full and would be entirely blue, so they are listed here instead.

1. What did not change

- Table 1, all ten tasks and every arm — the same generated file, every digit identical.
- Figure 1 — the same file, including the row shading added on 19 September.
- “improves the readout on 7 of 10 tasks”, and every number quoted in the abstract and §3.
- The reference list: same 22 entries, no additions.
- The abstract and §1, word for word. §2 changed in one place only, item 1 below.

2. Added to the body

1. **The rotation confound, disclosed and then answered.** Our first star pool admits a star as a positive only through the transit, eclipsing-binary or pulsating catalogues, and screens its quiet stars against the rotation catalogue as well. Every rotator in it therefore carries a second label, and no rotation-only star is in it — which is exactly the wrong pool on which to score rotation. §2 now says so, and §3 gives the two rotation tasks rescored on a third pool of 106,284 stars, the corpus outside the first pool, where rotators are a natural 13%. Neither verdict changes the count of 7: rotation gains $+0.055 \pm 0.006$ there, but an untrained encoder also gains $+0.029$, so about half of it is what any 128 extra columns buy; rotation period gains $+0.034 \pm 0.002$ with the untrained control at zero. The full measurement is new Appendix J.
2. **The validation-loss figure moved into the body** (now Figure 2), next to the paragraph that argues from it. It was Appendix C before, and a reader had to take the claim on trust.
3. **The within-window artifact, in one sentence in §4** pointing to new Appendix F: the objective develops a localised artifact inside each window, none of the interventions we tried cleared both pre-registered acceptance criteria, and the recipe we ship does so by the stated fallback rule rather than by winning.
4. **One sentence on where the nonlinear readout helps.** Where XGBoost raises the features-only score, the gain from adding μ falls; the rank correlation across the ten tasks is 0.92.
5. **Table 1’s caption now says what its bold means** — best point estimate, no significance test — because the appendix tables bold something else, a difference resolved beyond its error bar. The two conventions were silently different before.
6. **The discovery question is now marked as exploratory**, with the reason: with no labels there is nothing to score a latent-space separation against, so it needs a design of its own. And the reproducibility paragraph carries the actual repository link instead of “released on publication”.

3. Four new appendices

- **A — Pre-training and readout details.** What one pre-training example is, the first-difference term in the objective and why it is there, optimiser and schedule, batch size, epochs and patience, the beta warm-up and which checkpoint is frozen, standardisation using training-split statistics only, class weighting, and the settings of both supervised baselines.
- **B — The two star pools, and what a quiet star is.** The exact construction of both pools, and the statement that neither is screened against the other’s catalogues.
- **F — The within-window artifact, and the two gates that governed it.** Both pre-registered acceptance criteria verbatim, a reconstruction figure showing the artifact, the interventions we tried along four axes, and why the tapered recipe is the one we ship.
- **J — The two rotation tasks on a pool that contains rotation-only stars.** The measurement behind item 1 above, with the old and new numbers side by side.

4. Four appendices extended

- **C** gains “Why 128 redundant columns cost a linear fit”: the stars-per-coefficient argument for the two tasks where fusion loses, and the prediction it makes — an untrained encoder’s 128 columns should cost about the same as a trained one’s, and they do.
- **D** gains a per-task *retained* column in its table: the share of each gain that survives removing the GRU prediction term. The other columns are unchanged.
- **E** is split — its panel (a) is now Figure 2 in the body, panel (b) stays here — and says plainly that on the family our encoder belongs to the anti-correlation holds on two tasks and reverses on two, while the loss optimum is still never the probe optimum.
- **H** gains “What ‘modelled on T’DA’ means”: what we kept, changed and dropped relative to Audenaert et al. (2021), feature by feature.

5. Format

- Author block de-anonymised; first-page notice reads “Preprint.” instead of the workshop line.
- The three float-spacing overrides are gone. They existed only to hold four pages.
- The paper is 15 pages: body through page 5, references, then ten appendices.

Appendix letters moved, because four were inserted. The marked-up pages use the *arXiv* version’s letters throughout, including inside deleted text.

ML4PS	arXiv	
—	A	Pre-training and readout details (new)
—	B	The two star pools, and what a quiet star is (new)
A	C	The two tasks where fusion does not help
B	D	The gain under a nonlinear readout
C	E	Validation reconstruction against downstream score
—	F	The within-window artifact, and the two gates (new)
D	G	Paired bootstrap over test stars
E	H	The 25 engineered features
F	I	Evaluation restricted to stars never seen in pre-training
—	J	The two rotation tasks on a rotation-only pool (new)

The marked-up body starts on the next page.

What does self-supervision add to engineered features for stellar light curves?

Anonymous Author(s) Zijun Liu
Affiliation [University of California, San Diego](#)
Address brl1055@ucsd.edu
email liuzijun6688@gmail.com

Yue Ma
[Halicioğlu Data Science Institute](#)
[University of California, San Diego](#)

Christopher A. Theissen
[Department of Astronomy and Astrophysics](#)
[University of California, San Diego](#)

Abstract

TESS light curves encode a star’s pulsation, rotation, binarity and oscillations, among other physical properties, but the survey produces far more light curves than can be labelled by hand. Astronomers have engineered a few dozen physically motivated features from them, which together remain a strong baseline. Previous work has trained encoders on light curves and judged them by their own downstream scores, but has not asked what they add to the engineered features. We pre-train a convolutional encoder on unlabelled TESS light curves to predict each window’s latent representation from those of its neighbours. We then freeze it, add its features to the 25 engineered features, and read the combination out with one fixed linear probe on ten tasks: five variability classes, four asteroseismic targets and rotation period. Adding the learned features improves the readout on 7 of 10 tasks over the engineered features alone, while an untrained encoder does not. The combination also outperforms two networks trained end-to-end on the same labelled stars on most tasks. An encoder trained without labels therefore gives astronomers new features that complement the ones they already use rather than replace them.

1 Introduction

A star’s light curve carries its pulsation, rotation, eclipses and oscillations, and with them its age, interior and companions. Wide-field surveys now produce light curves far faster than they can be labelled. TESS [Ricker et al., 2015] returns 2-minute or 20-second [Huber et al., 2022] light curves for hundreds of thousands of pre-selected stars, and full-frame images of every star in the field at cadences that change between mission cycles (30 minutes, then 10, now 200 seconds). LSST [Ivezić et al., 2019] will add *billions* more light curves, albeit at a cadence of days. Self-supervised learning is the natural response: pre-train one encoder on the unlabelled archive, then attach a cheap readout for each task. Encoders of this shape exist for ground-based, Gaia and Kepler light curves [Donoso-Oliva et al., 2025, Zuo et al., 2025, Rui, 2026] and are evaluated by classification accuracy, sometimes against engineered features [Rui, 2026]; a label-free TESS representation [Poznanski, 2026] is evaluated by whether repeat observations of one star lie close in its embedding space. None measures what they add to engineered features.

In this domain, the baseline that a learned encoder must improve on is already strong. Decades of variable-star research have distilled light curves into compact engineered summaries: periodogram peak structure, band-limited power, spectral entropy and robust scatter statistics. Pipelines built on them remain highly competitive. The benchmark of Li et al. [2025] makes the point sharply:

across seven expert-labelled variable-star classes, no time-series foundation model they tested strictly surpassed the long-established domain feature baseline on classification. A learned encoder for this data therefore has to answer one question: do its features add predictive signal to the engineered features under the same readout?

We measure that directly. We pre-train an encoder on TESS light curves, freeze it, and add its features to the 25 engineered features under one fixed linear readout. We ask two things: what the learned features add, and whether the combination, which trains only a linear model on the labels, outperforms two networks trained end-to-end on the same labels. Our contributions are:

- **The learned features add to the engineered features rather than replacing them.** Adding the learned features to the engineered ones improves a fixed linear readout beyond 2 SE on 7 of 10 tasks; an untrained encoder gives no such gain, so the gain comes from pre-training, not from merely adding 128 columns.
- **The combined pipeline, read out by a fixed linear model, outperforms the two supervised networks we train end-to-end on the same labels on most tasks.** Against a supervised Conv1D with the encoder’s architecture it is ahead on 6 of 10 tasks and indistinguishable on the other 4; against an MLP on raw flux it is ahead on 9 of 10.

2 Data and method

Data. We use 2-minute SPOC PDCSAP light curves of stars brighter than TESS magnitude $T = 10$ from Sectors 27–101. Pre-training and five of the ten probes (pulsating, eclipsing binary, rotation, transit and rotation period) use a label-enriched subset of 13,470 stars, split 70/15/15 by star: every star labelled as a transit host, eclipsing binary or pulsator that has a usable window, plus a random sample of 10,000 quiet stars, listed in none of our variability catalogues. The pre-training loss never sees a label, but this pool is label-enriched, not a random draw from the survey. The other five probes draw their labels from catalogues that barely overlap this subset and are scored on a second pool of 22,860 stars built the same way (every catalogue positive with a usable window plus 10,000 quiet stars, split 70/15/15). No test star of the first pool enters pre-training. The first pool admits positives only through the transit, eclipsing-binary and pulsating catalogues and screens its quiet stars against the rotation catalogue as well, so every rotator in it carries a second label and it holds no rotation-only star; Appendix J rescues the two rotation tasks on a third pool of 106,284 stars, the corpus outside the first pool, where rotators are a natural 13%. About 8% of the second pool’s test stars (265 of 3,429) were in the pre-training subset; removing them leaves all seven gains in place (Appendix I). Each light curve is split into contiguous segments wherever the time axis breaks, at a missing cadence or a jump longer than five median cadences; segments are median-normalised by their median absolute deviation and sliced into non-overlapping windows of 256 cadences ($256 \times 120 \text{ s} \approx 8.5 \text{ h}$); restricting to one cadence keeps every window the same span of time. Windows containing a missing value are discarded, never interpolated or padded, and segments are never stitched across sector or gap boundaries. Pre-training consumes sequences of exactly 16 consecutive windows ($\approx 5.69 \text{ d}$), cut at a random offset within a segment, and uses no labels.

Encoder. A 1D convolutional variational encoder [Kingma and Welling, 2014] maps each window to a 128-dimensional Gaussian posterior, and a GRU [Cho et al., 2014] predicts the next window’s latent from its predecessors, run both forward and backward over the sequence. The objective is

$$\mathcal{L} = \|\hat{x} - x\|^2 + \beta \text{KL} + \lambda [\|\hat{z}_{t+1} - \bar{z}_{t+1}\|^2 + \|\hat{z}_{t-1} - \bar{z}_{t-1}\|^2] \\ + w [\|\log P_H(\hat{x}) - \log P_H(x)\|^2 + \|\Delta\hat{x} - \Delta x\|^2],$$

where every term is a mean squared error, $\bar{z}$ the stop-gradient target of the forward and backward predictors, P_H the Hann-tapered power spectrum, Δ the first difference, and $(\beta, \lambda, w) = (0.1, 60, 0.3)$, β warmed up linearly over ten epochs. The encoder is trained with six seeds and then frozen; no gradient from any label reaches it.

Representations and readout. A star’s learned features are μ , the mean, over every window of its first segment (16 for 95% of stars, 20 for the rest), of the per-window posterior means. The engineered baseline is a fixed 25-feature vector modelled on the T’DA variability pipeline [Audenaert et al., 2021], computed on the same first segment, 19 from its periodogram and 6 time-domain statistics;

Table 1: Ten-task scorecard. Frozen arms: a fixed logistic or ridge readout, or XGBoost on the same inputs (Appendix D). Bold: best point estimate of the three arms under each frozen readout, with no significance test attached; bold in the appendix tables instead marks a difference resolved beyond its error bar, so the two conventions are not interchangeable. n : test stars; prev.: positive-class prevalence.

task	metric	n (prev.)	frozen linear readout			frozen XGBoost readout			end-to-end supervised	
			features	μ	features $\oplus \mu$	features	μ	features $\oplus \mu$	Conv1D	MLP
pulsating	PR-AUC	2021 (0.107)	0.789	0.806	0.846	0.845	0.821	0.834	0.816	0.748
eclipsing binary	PR-AUC	2021 (0.097)	0.742	0.771	0.775	0.798	0.789	0.807	0.774	0.719
rotation	PR-AUC	2021 (0.089)	0.540	0.559	0.555	0.596	0.544	0.576	0.565	0.526
transit	PR-AUC	2021 (0.060)	0.190	0.144	0.218	0.217	0.223	0.228	0.134	0.109
oscillating giant	PR-AUC	3429 (0.383)	0.920	0.854	0.918	0.924	0.899	0.923	0.875	0.883
solar-like osc.	PR-AUC	3429 (0.122)	0.320	0.336	0.390	0.470	0.409	0.463	0.396	0.275
RGB vs. HeB	ROC-AUC	161 (0.702)	0.758	0.660	0.708	0.746	0.725	0.763	0.530	0.762
$\nu_{\max}$	R^2	1313	0.831	0.802	0.867	0.916	0.831	0.919	0.832	0.834
rotation period	R^2	150	0.703	0.677	0.717	0.695	0.655	0.730	0.597	0.698
flare activity	PR-AUC	3429 (0.089)	0.474	0.453	0.524	0.562	0.485	0.565	0.533	0.444

Appendix H defines each feature. Readouts are fixed in advance and never tuned on a task, with inputs standardised using training-split statistics alone: class-balanced L_2 logistic regression at $C = 1$ for the detection tasks, ridge for the two regressions with its penalty set by leave-one-out cross-validation. We compare four arms, each the same readout on a different feature set: `features`, μ , `features  $\oplus \mu$` , and, as a control, the same fusion built on an *untrained* encoder; the two supervised baselines below are the fifth and sixth arms. Appendix D repeats the first three under XGBoost.

Supervised baselines. Two end-to-end baselines see exactly the same input windows, star sets and splits. **C1** reuses the encoder’s convolutional trunk unchanged, adds a 128-unit linear layer and a scalar output, and is trained per task with a class-weighted loss on a star-level score, the mean of the star’s window logits computed inside the network so that the gradient flows through the mean. **C2** replaces the convolutional trunk with a dense one and changes nothing else, so C1 versus C2 isolates the convolutional inductive bias. Both use one pre-registered regularisation setting for all tasks, select on a validation split no probe has ever touched, and are trained with six seeds.

Tasks and metrics. We score ten tasks with one probe per physical quantity: pulsating variables [Gao et al., 2025], eclipsing binaries [Prša et al., 2022], rotation and rotation period [Boyle et al., 2026], transits [Guerrero et al., 2021], flare activity [Seli et al., 2025, Vida et al., 2021], oscillating red giants and their $\nu_{\max}$ [Hon et al., 2021], solar-like oscillators [Hatt et al., 2023], and RGB versus helium-burning discrimination [Sreenivas et al., 2026]. Here $\nu_{\max}$ is the frequency of maximum oscillation power. Detection tasks are scored by the area under the precision–recall curve (PR-AUC), whose floor is the positive-class prevalence rather than 0.5; prevalence is therefore quoted with every detection score. The two regressions are scored by R^2 .

Uncertainties. All error bars are $2\times$ the standard error of the mean over seeds. For the μ arms the seeds are independent pre-training runs (six); for C1 and C2 they are initialisation and shuffling seeds (six); the engineered arm is deterministic. The seeds of the μ arms and of the supervised arms are unrelated, so differences between them are treated as unpaired and their bands combined as $\sqrt{\text{SE}_a^2 + \text{SE}_b^2}$. A difference counts only when it exceeds the combined 2 SE band. These bands measure re-training spread only; a paired bootstrap over the test stars (1,000 resamples, seeds averaged; Appendix G) checks that no verdict hinges on the particular test stars.

3 Results

The learned features add to the engineered features. Table 1 and Figure 1a give the comparison. Adding μ to the 25 features improves the fixed readout beyond 2 SE on 7 of 10 tasks, led by solar-like oscillators $+0.071$, pulsating $+0.057$ and flare activity $+0.051$, and reaching $+0.013$ (R^2) on rotation period. Rotation is positive but just short of the bar ($+0.016$ against 0.016). On the rotation pool of Appendix J, where the two classes differ in rotation and nothing else, the rotation gain is $+0.055 \pm 0.006$ but an untrained encoder also gains $+0.029$, so about half of it is what any 128

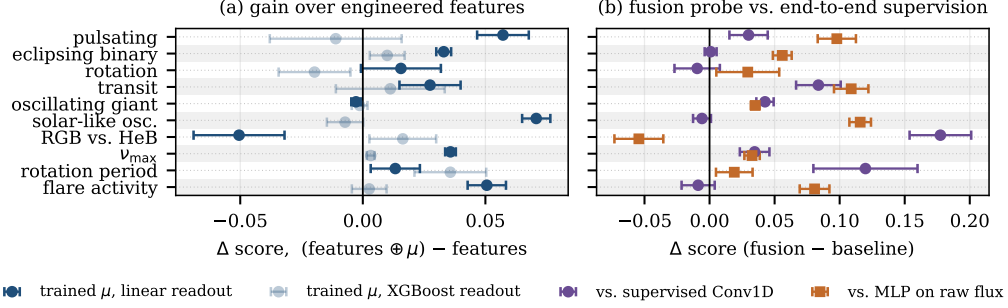


Figure 1: The two claims. **(a)** What the learned features add to the engineered features, under the fixed linear readout of Table 1 (solid) and under an XGBoost readout on the same inputs (faded; Appendix D). **(b)** The fusion probe against the two supervised baselines. Bars are 2 SE.

extra columns buy on 74,000 training stars; the rotation-period gain there is $+0.034 \pm 0.002$ with the untrained control at zero. Neither verdict moves the count. The two negative tasks are the ones where 128 extra columns cost more than they add: the 25 features already cover the task (oscillating giants, -0.003 on a baseline of 0.920) or the training set is too small to fit 128 more weights (RGB versus helium-burning, -0.050 on 755 training stars). On both, an untrained encoder loses by about the same amount as the trained fusion, within the error band; that is what redundant columns do to a linear fit, not a sign that μ carries misleading content (Appendix C). Across all ten tasks the untrained control is negative on seven and never exceeds $+0.016$ (Table D.1), so the gains are not what any 128 extra columns would buy on this feature set; they are specific to what pre-training learned. μ on its own does not replace the features (ahead on only 4 of 10 tasks alone), as Li et al. [2025] also find for time-series foundation models; we therefore report fusion rather than substitution [Makinen et al., 2024].

The fusion probe outperforms the two supervised networks on most tasks. Against C1, which shares the convolutional trunk, labels and input windows but is trained end-to-end, the fusion probe is ahead on 6 of 10 tasks and indistinguishable from it on the other 4 (Figure 1b). Against C2, an MLP on raw flux, it is ahead on 9 of 10 and behind on one (RGB versus helium-burning, 161 stars). C1 is not weak: it is ahead of the engineered arm on 5 of 10. A paired star-bootstrap (Appendix G) excludes zero on 5 of the 7 fusion gains, 5 of the 6 leads over C1 and 7 of the 9 over C2. This holds for this architecture on 669 to 16,002 labelled stars per task, not for supervision in general.

The GRU prediction term raises the gain on every task. Removing the GRU prediction term, all else fixed and the same six seeds, reduces the fusion gain on all 10 tasks, and by more than 2 SE on 6 when the two arms are differenced seed by seed (Table D.1). Without it the gain falls by more than half on 5 of the 8 tasks where fusion gains.

4 Discussion

Pick the encoder by a held-out probe, not by its loss. Across the ten pre-training recipes of one sweep, a lower validation reconstruction loss went with a *lower* probe score: the Spearman ρ between negated loss and PR-AUC runs from -0.58 to -0.84 over four tasks (Figure 2). The recipe with the lowest loss is the one without the prediction term on all four, and choosing it would cost 0.021–0.049 PR-AUC. On the family our encoder belongs to the anti-correlation does not survive as such — it holds on two of the four tasks and reverses on two — but the loss optimum is still never the probe optimum (Appendix E). The same objective carries a second cost: it develops a localised artifact inside each window, and none of the interventions we tried on four axes cleared both pre-registered acceptance criteria, so the recipe we ship does so by the stated fallback rule rather than by winning (Appendix F). An encoder here therefore has to be chosen by a held-out probe, not by its own loss.

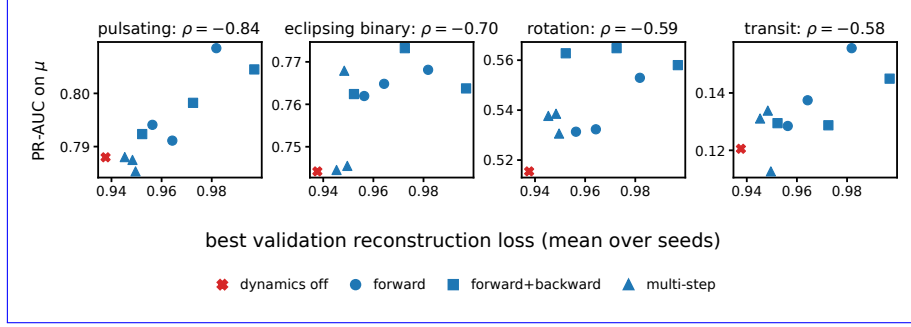


Figure 2: Validation reconstruction loss against downstream probe score, one point per pre-training recipe and one panel per task. The ten recipes share corpus, architecture and loss terms and differ only in the GRU prediction term (off, forward, forward+backward, multi-step, the last three at three weights), four seeds each. x : best post-warm-up validation reconstruction loss, averaged over the recipe’s seeds; y : PR-AUC of the fixed linear probe on μ ; ρ : Spearman’s rank correlation between negated loss and score. The loss-optimal recipe is the dynamics-free one (red) on all four tasks, and it is the worst or next-to-worst probe on each. Appendix E repeats the measurement on the family this paper’s encoder belongs to.

Limitations and next steps. *Readout.* The four linear-readout arms (features, μ , features $\oplus \mu$, and the untrained control) share one linear model. It has few parameters, so it cannot rescue weak features, and it is identical across the four, so the comparison is between feature sets. Our claim is confined to these 25 features and a linear readout: under XGBoost on the same inputs the gain exceeds 2 SE on 4 of 10 tasks and is negative on one; where the nonlinear readout raises the features-only score, the μ gain falls, with a rank correlation of 0.92 across the ten tasks (Appendix D). *Data.* Every arm, the supervised baselines included, sees one 5.7-day segment of a bright 2-minute star. The fainter, far more numerous stars in the full-frame images, and light curves spanning several sectors, are the obvious next experiments. *Labels.* Our negatives are stars absent from the label catalogues, not stars verified to lack the signal, so catalogue incompleteness adds positive-unlabelled label noise and the detection scores may be conservative. The flare label marks stars that have flared somewhere in their TESS data, not a flare inside the 5.7-day segment we encode, so the task is recognising a flaring star, not finding a flare. *Label scarcity.* When the readout is trained on a few percent of the labels, the advantage narrows on 7 of 10 tasks, so how much pre-training helps when labels are scarce remains open. *Discovery.* Every task here has a catalogue; whether μ also separates kinds of variability that no catalogue labels yet, for instance as clusters in the latent space, is untested. We leave it as an exploratory direction rather than a result: with no labels there is nothing to score such a separation against, so it needs a design of its own.

Reproducibility and disclosure. §2 and Appendix H give the data, objective, features and readouts; the encoder has four convolutional stages (widths 32/64/128/256, kernel 5, each halving length), a 128-d latent and a one-layer GRU of width 256. Code and score tables will be released on publication, configuration files and the score tables behind Table 1 and the appendix tables are available at <https://github.com/BrightonLiu-zz/stellar-ssl/tree/arxiv-v1>; generative AI assisted with analysis code and prose editing.

References

- Jeroen Audenaert, James S. Kuszlewicz, Rasmus Handberg, et al. TESS data for asteroseismology (T²DA) stellar variability classification pipeline: Set-up and application to the Kepler Q9 data. *The Astronomical Journal*, 162(5):209, 2021. arXiv:2107.06301.
- Andrew W. Boyle, Luke G. Bouma, and Andrew W. Mann. The TESS all-sky rotation survey: Periods for 1,046,317 stars within 500 pc. *arXiv e-prints*, 2026. arXiv:2603.05586.
- Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, et al. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In *Conference on Empirical Methods in Natural Language Processing*, 2014. arXiv:1406.1078.
- Cristobal Donoso-Oliva, Ignacio Becker, Pavlos Protopapas, et al. Astromer 2. *arXiv e-prints*, 2025. arXiv:2502.02717.
- Xinyi Gao, Xiaodian Chen, Shu Wang, and Jifeng Liu. Classification of periodic variable stars from TESS. *The Astrophysical Journal Supplement Series*, 276(2):57, 2025. arXiv:2412.06175.
- Natalia M. Guerrero, Sara Seager, Chelsea X. Huang, et al. The TESS objects of interest catalog from the TESS prime mission. *The Astrophysical Journal Supplement Series*, 254(2):39, 2021. arXiv:2103.12538.
- Emily Hatt, Martin B. Nielsen, William J. Chaplin, et al. A catalogue of solar-like oscillators observed by TESS in 120-second and 20-second cadence. *Astronomy & Astrophysics*, 669:A67, 2023. arXiv:2210.09109.
- Marc Hon, Daniel Huber, James S. Kuszlewicz, et al. A ‘quick look’ at all-sky galactic archeology with TESS: 158,000 oscillating red giants from the MIT quick-look pipeline. *The Astrophysical Journal*, 919(2):131, 2021. arXiv:2108.01241.
- Daniel Huber, Timothy R. White, Travis S. Metcalfe, et al. A 20-second cadence view of solar-type stars and their planets with TESS: Asteroseismology of solar analogs and a re-characterization of π Men c. *The Astronomical Journal*, 163(2):79, 2022. arXiv:2108.09109.
- Željko Ivezić, Steven M. Kahn, J. Anthony Tyson, et al. LSST: From science drivers to reference design and anticipated data products. *The Astrophysical Journal*, 873(2):111, 2019. arXiv:0805.2366.
- Diederik P. Kingma and Max Welling. Auto-encoding variational bayes. In *International Conference on Learning Representations*, 2014. arXiv:1312.6114.
- Weijian Li, Hong-Yu Chen, Nabeel Rehemtulla, et al. StarEmbed: Benchmarking time series foundation models on astronomical observations of variable stars. *arXiv e-prints*, 2025. arXiv:2510.06200.
- T. Lucas Makinen, Ce Sui, Benjamin D. Wandelt, et al. Hybrid summary statistics. *arXiv e-prints*, 2024. arXiv:2410.07548.
- Dovi Poznanski. A useful representation of TESS light curves. *arXiv e-prints*, 2026. arXiv:2605.05324.
- Andrej Prša, Angela Kochoska, Kyle E. Conroy, et al. TESS eclipsing binary stars. I. short-cadence observations of 4584 eclipsing binaries in sectors 1–26. *The Astrophysical Journal Supplement Series*, 258(1):16, 2022. arXiv:2110.13382.
- George R. Ricker, Joshua N. Winn, Roland Vanderspek, et al. The transiting exoplanet survey satellite. *Journal of Astronomical Telescopes, Instruments, and Systems*, 1(1):014003, 2015. arXiv:1406.0151.
- Yicheng Rui. Domain-informed multi-view self-distillation for astronomical light-curve representation learning with JEPA. *arXiv e-prints*, 2026. arXiv:2606.28446.
- Bálint Seli, Krisztián Vida, Katalin Oláh, et al. Stellar flare morphology with TESS across the main sequence. *Astronomy & Astrophysics*, 694:A161, 2025. arXiv:2412.12989.

- K. R. Sreenivas et al. Global asteroseismology of 19,000 red giants in the TESS continuous viewing zones. *arXiv e-prints*, 2026. arXiv:2604.00498.
- Krisztián Vida, Attila Bódi, Tamás Szklénár, and Bálint Seli. Finding flares in Kepler and TESS data with recurrent deep neural networks. *Astronomy & Astrophysics*, 652:A107, 2021. arXiv:2105.11485.
- Xiaoxiong Zuo, Yihan Tao, Yang Huang, et al. FALCO: A foundation model of astronomical light curves for time-domain astronomy. *arXiv e-prints*, 2025. arXiv:2504.20290.